

Practice Quiz: Learning (Neuroscientific Frontiers)

Name: __ Date: __

Part A: Multiple Choice

1. LTP implements which Active Inference operation? A) Action selection B) Strengthening association parameters in the generative model — increasing the likelihood of observed co-occurrences C) Perception D) Sleep
2. STDP implements causal inference because: A) All plasticity is causal B) Pre-before-post timing (causal direction) produces LTP, while post-before-pre (acausal) produces LTD — learning temporal causation C) Spikes don't have timing D) Only inhibitory neurons use STDP
3. Memory consolidation involves: A) Permanent hippocampal storage B) Transfer of parameters from hippocampus (fast learning) to neocortex (slow integration) through sleep-dependent replay C) Forgetting everything D) Only cortical processing
4. Sharp wave-ripples implement: A) Dream content B) Compressed replay of waking experiences, coordinated with cortical slow oscillations and spindles to facilitate consolidation C) Pain signals D) Motor commands
5. Synaptic pruning during development implements: A) Brain damage B) Bayesian Model Reduction — removing synapses that contribute little to model evidence, creating a more efficient generative model C) Random loss D) Growth
6. Adult hippocampal neurogenesis may implement: A) Memory erasure B) Structure learning — adding new representational capacity to encode novel contexts and environments C) Emotion D) Language
7. The precision of prediction errors modulates learning rate through: A) Blood flow B) Neuromodulation — high-precision errors (salient, reliable) produce faster learning; low-precision errors produce slower learning C) Skull thickness D) Age alone

Part B: Essay Questions

1. Compare three theories of sleep function: (a) synaptic homeostasis hypothesis (Tononi & Cirelli), (b) active systems consolidation (Born & Wilhelm), and (c) Active Inference's BMR interpretation. What does each theory predict about what happens during NREM vs. REM sleep? Can they be unified? (400 words)
2. A patient with bilateral hippocampal damage (like patient HM) can no longer form new declarative memories but can still learn motor skills. Explain this dissociation using the Active Inference learning framework. Which learning mechanisms are preserved? Which are lost? How does this support the complementary learning systems view? (400 words)
3. Design a computational model that implements Active Inference learning using biologically realistic plasticity rules. Specify: (a) what the generative model represents, (b) which plasticity rule updates which parameters, (c) how prediction errors drive learning, and (d) how a sleep-like consolidation phase would work. This can be schematic/theoretical — no code required, but be specific about the computational operations. (400 words)